Blood Flow Analysis Software Based On Duplex Doppler Ultrasound Recordings

Team 12

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User Needs and Project Statement

The original information from Progress Report 1 remains current. The problem statement is as follows.

Taha Abassi-Hashemi and the labs he leads need a robust and efficient software system capable of batch processing duplex Doppler ultrasound recordings to extract reliable vascular geometry and blood flow measurements, while enabling human validation and integration with existing research workflows. Previous ultrasound processing software is outdated and requires unnecessary researcher intervention, as well as lacks crucial/accurate metrics that the lab requires for research. These factors cause the lab to have missing and inaccurate data.

Design Inputs

The following are the user needs are listed in Progress Report 1:

1. Accurate characterization of vessel geometry
2. Extraction of blood-flow information from Doppler recordings
3. Human review and correction of automated measurements
4. Integration with existing research workflows

These user needs along with safety, legal, and regulatory inputs are mapped onto requirements. The concept of constraints and objectives as defined by [1] align with the software definition of functional and nonfunctional requirements and will be mapped as such. Functional requirements are denoted by F.x.x. and non-functional requirements are denoted by NF.x.x.

Requirements

Foundational Image Processing Requirements

F.1.1.	The software shall reliably segment ultrasound video frames to isolate the B-mode image, the Doppler graph, and the scale areas dynamically.
F.1.2.	The system shall isolate the depth scale, the actual ultrasound image, and theh placement vector from the B-mode region.
F.1.3.	The system shall separate the green ECG trace from the white Doppler waveform within the Doppler region.
F.1.4.	The system shall read the numerical values of the depth and velocity scales to obtain the scale of the images
F.1.5.	The system shall track the recording timeline
F.1.6.	The system shall accommodate specific ultrasound views where the Doppler waveform is entirely absent.
F.1.7.	The system shall locate the on-screen bounding green "Z" box to establish the initial region of interest and utilize a breadth-first search to locate the surrounding vessel walls.

NF.1.6.	The software shall handle data interruptions
NF.1.7.	The system shall maintain processing continuity if the depth or scale changes dynamically mid-recording
NF.1.8.	The system shall accommodate recordings where the time scale changes dynamically during the video.
NF.1.9.	The system shall maintain tracking continuity when unexpected physical artifacts occur during the ultrasound recording, such as patient coughing, laughing, or hyperventilating.

Accurate Characterization of Vessel Geometry

F.2.1.	The software shall accurately extract vessel diameter perpendicular to the vascular walls in longitudinal views
F.2.2.	The system shall calculate the metrics for multiple vessel candidates in the B-mode view to determine the best candidate for tracking
F.2.3.	The system shall determine the cross-sectional area of circular vessels in transverse views using edge detection

Extraction of Blood-Flow Information

F.3.1.	The system shall extract time-average blood velocity from the Doppler waveform.
F.3.2.	The system shall calculate estimated volumetric blood flow using the extracted velocity and calculated diameter.

Human Review and Correction

F.4.1.	The software shall provide an interactive GUI that allows human review and manual correction of automated edge-detection measurements.
F.4.2.	The GUI shall project an augmented visual overlay directly on top of the original ultrasound video to display the automated edge detection.
F.4.3.	The GUI shall support distinct user states, specifically allowing the user to process, edit, pause, cancel, and undo actions.
F.4.4.	The system shall implement a revision history that permanently preserves the original automated machine result whenever a user inputs a manual edit.

Integration with Existing Research Workflows

F.5.1.	The system shall integrate directly with version 2 of PhysioMerge by functioning as a package that PhysioMerge can call via Command Line Interface (CLI) arguments.
F.5.2.	The software shall include a test harness capable of running the different image processing tasks against a test dataset with an established ground truth.
F.5.3.	The application shall be designed to operate within the established hardware constraints for CPU or GPU real-time processing.
F.5.4.	The system shall output a csv file that includes all the extracted and calculated metrics.
NF.5.5.	The system shall efficiently manage large file sizes, accommodating constant frame rate video recordings that can reach 6 GB for a 30-minute session.
NF.5.6.	The application shall be optimized to run on an NVIDIA RTX A5000 GPU and an Intel Xeon w-2245 CPU with 32 GB of RAM, while including a fallback processing mode for less powerful machines.

Regulatory, Legal & Ethical Inputs

NF.6.1.	The software architecture shall utilize the IEEE 754 standard for floating-point arithmetic to prevent numerical inaccuracies during data conversion.
NF.6.2.	The software shall align with Medical Devices Regulations (SOR/98-282) as an "active diagnostic device" and IEC 62304 for Software as a Medical Device.
NF.6.3.	The design shall comply with PIPEDA and PHIPA privacy regulations regarding identifiable health records, such as combined ECG and ultrasound data.
NF.6.4.	The system shall utilize dependencies with Open Source Software (OSS) licenses that are suitable for the project's requirements.
NF.6.5.	The design shall address ethical concerns regarding over-reliance on automation to ensure clinicians retain their skills in manually verifying data.

Usability & UI Nonfunctional Requirements

NF.7.1.	The GUI shall remain responsive to user inputs and window scaling without freezing during heavy continuous edge-detection workloads.
NF.7.2.	The software shall initialize and present a ready-state GUI within 5 seconds of launch on standard lab desktop hardware.
NF.7.3.	The system shall provide clear, non-technical error notifications when a user uploads an unsupported video format or a corrupted file.
NF.7.4.	The application shall adopt a minimalist, high-contrast visual design that ensures the augmented edge-detection overlay is clearly visible against grayscale B-mode ultrasound backgrounds.

Performance & Efficiency Nonfunctional Requirements

NF.8.1.	The system shall process single ultrasound video files at a rate equal to or faster than the original video playback duration (i.e., minimum 1x processing speed).
NF.8.2.	The application's memory footprint shall be optimized to not exceed standard desktop RAM constraints (e.g., 8GB) during batch processing to prevent system crashes.
NF.8.3.	The software shall parse and extract initial metadata (duration, resolution, frame rate) from a selected directory of batched videos within 10 seconds.

Reliability & Robustness Nonfunctional Requirements

NF.9.1.	The software shall fail gracefully by logging the error and skipping to the next file without crashing the entire application if a single video in a batch queue is corrupted.
NF.9.2.	The system shall maintain consistent edge detection and UI overlay accuracy across various standard display resolutions and aspect ratios.
NF.9.3.	The application state and user-defined region of interest (ROI) parameters shall be temporarily cached so data is not lost if the user accidentally closes the active processing window.

Maintainability & Extensibility Nonfunctional Requirements

NF.10.1.	The software architecture shall be modularly designed to support future advanced signal processing extensions, such as Intima-Media Thickness (IMT) calculations.
NF.10.2.	The codebase shall be thoroughly documented internally to facilitate seamless handoffs and continued development by future researchers in the Vascular AI lab.

NF.10.3.	The application shall use a standardized configuration file structure for its parameters to allow researchers to update default values without altering the core source code.

Non-functional Requirement/Design Input Development with Stakeholder Consultation

The majority of the requirements are derived from past Progress Reports and initial stakeholder consultation, while the following requirements were traced from the most recent one from October 5th, 2026 [2], where Taha performed another FMD recording (this time with common issues and an intervention) as well as showed us how he uses the PhysioMerge software.

- F.1.6: Support for views without Doppler waveforms.
- F.1.7: Use of the green "Z" box and BFS algorithm for bounding.
- F.4.4: Revision history preserving original machine results alongside human edits. NF.1.8: Handling of dynamic time scale changes mid-recording.
- NF.1.9: Resilience to physical artifacts (e.g., patient coughing, laughing).
- F.5.1: PhysioMerge v2 integration through CLI arguments.
- NF.5.5: Capacity for 6 GB file sizes per 30-minute recording.
- NF.5.6: Specific workstation constraints (NVIDIA RTX A5000, Xeon w-2245, 32GB RAM).

Design Input Mapping

The requirements (formalized from the user needs) are mapped into design inputs shown in Figure 1. Specific requirements were grouped together on the criteria of proximity to avoid repetitive and excess design inputs.

Project:	Blood Flow Analysis Software Based On Duplex Doppler Ultrasound Recordings										
Date:	Oct-26										
Relative Weight	Customer Importance 1-5 (1 = low, 5 = High)	Customer Requirements	The software shall segment B-mode, Doppler, and scale regions with >95% accuracy on standard lab datasets.	The system shall automatically calibrate physical pixel scale from the on-screen rulers to within 1mm accuracy.	The system shall measure perpendicular vessel diameter and cross-sectional area with <5% error compared to clinical baselines.	The software shall extract time-averaged blood velocity and calculate estimated volumetric flow rate in mL/min.	The GUI shall overlay an interactive edge-detection spline on the ultrasound video with system latency under 100ms.	The system shall export numerical output directly to a PhysioMerge-compatible CSV format containing all 7 target metrics.	The software shall batch process a queue of 50 consecutive 1080p ultrasound videos at a minimum of 1x real-time playback speed.	The architecture shall utilize IEEE 754 floating-point arithmetic to prevent truncation errors during data conversion.	The software shall encrypt combined patient ECG and ultrasound data to comply with PIPEDA and PHIPA regulations.
13%	5	Reliable isolation of B-mode, Doppler, and scale regions	5								
10%	4	Processing continuity during data interruptions or scale changes	3	3					3		
13%	5	Accurate characterization of vessel geometry	3	5	5		3				
13%	5	Extraction of blood-flow information	3	5		5	1				
13%	5	Interactive human review and manual correction			3		5				
10%	4	Integration with PhysioMerge and existing workflows						5			
10%	4	Efficient batch processing of recordings	3	3	3	3			5		
13%	5	Compliance with medical device and privacy regulations								3	5
8%	3	Ethical design to prevent clinician over-reliance on automation					5				
0%											
0%											
0%											
0%											
0%											
0%											
		Importance Rating Sum (Importance x Relationship)	1.975	1.6	1.675	0.675	1.75	0.875	1.175	0.75	1
		Relative Weight	17%	14%	15%	6%	15%	8%	10%	7%	9%

Correlations	
Positive	+
Negative	-
No Correlation	

Relationships	Weight
Strong	5
Medium	3
Weak	1

Figure 1. Requirement to Design Input Mapping.

Concept Development Process and Preliminary Analysis

Concept Generation and Feasibility Analysis

To systematically generate, analyze, and compare design concepts for the blood flow analysis software, an iterative, component-based development cycle is employed. The structured process for the fall semester consists of splitting the various inputs from the duplex Doppler ultrasound videos, researching the underlying physiological and technical mechanisms for each, and then iteratively building and testing throwaway prototypes for those functional parts. This ensures that high-risk components are validated for feasibility before final system integration.

Excalidraw was used for the breakdown of the component-based development cycle in order to inform further work as well as task dependencies. It is attached in Appendix A.

Uncertainties and Trade-offs

During the concept development and selection process, several trade-offs and constraints must be weighed:

- **Hardware and Performance Constraints**: The system must balance the computational load of continuous edge detection and video processing against standard hardware

limitations. We are evaluating trade-offs between CPU and GPU real-time processing capabilities to ensure the software remains responsive.

- **Legal and Technical Trade-offs**: As the software will rely on external libraries for numerical analysis and image processing, we must analyze the permissiveness of various Open-Source Software (OSS) licenses to ensure compliance with our project requirements.
- **Ethical and Societal Factors in Automation**: A major trade-off in medical software is balancing automated algorithmic efficiency with the risk of user over-reliance. Shifting analysis heavily onto machines introduces an ethical concern where reviewers may lose fundamental checking skills and blindly trust the automated output, potentially missing critical measurement inaccuracies. To mitigate this, our concepts emphasize a human-in-the-loop workflow, ensuring the user can pause, edit, and adjust automated regions of interest.

Preliminary Design Configuration and Testing

Concepts will be iteratively refined by building a robust test harness to evaluate early throwaway tests against expected outputs. Viable components will be consolidated into a preliminary Figma UI overlay, defining states for processing, editing, pausing, and undoing actions. These elements will ultimately be tested using preliminary flow-mediated dilation (FMD) imaging data to verify integration with the vascular lab's existing PhysioMerge software. Data under different conditions is being gathered during the weekly stakeholder consultations.

Preliminary Risk Analysis

Early Health and Safety Risks

Health and safety risks for this project are entirely found within the data collection stage. As a software-based solution for analyzing doppler ultrasound videos, there exists no such risks in any other stages as the only user is the operator and they are analyzing videos on a screen. Data collection is used for verification and validation – ensuring that the software accurately processes the videos and later ensuring that the outputs regarding the processed videos are what the user wants. Health and safety risks are involved because members of this group must take ultrasound recordings of themselves for testing purposes – the Vascular AI lab cannot provide patient data for privacy reasons.

Ultrasound recordings have a few health and safety risks. Firstly, allergic reactions can occur from the water-based ultrasound gel used in the procedure. These are minor and can be avoided by testing other group members. Secondly, some physical discomfort can be experienced when the operator (Taha) is maneuvering the ultrasound device to get a clearer picture of the blood vessels, such as when the neck has pressure applied to get a clearer view of the common carotid artery. Outside of these minor situations, no health and safety risks exist within data collection or anywhere else in the project.

Applicable Standards

Given that the software is involved in ultrasound recordings which may be used for clinical purposes, the "device" may fall under:

- Medical Devices Regulations (SOR/98-282) as an "active diagnostic device", as it is an image/video analysis tool that provides quantitative diagnostic outputs [3].
- IEC 62304 as a standalone "Software as a Medical Device" (SaMD). The device in research use falls under Class A with no injury or damage to health being possible, but if it is opened up to wider use in the future, it would fall under Class B due to non-serious injury risks from incorrect diagnostic data. This is relevant as it defines life cycle requirements, verification, and regressing tracking, and would focus on aspects of the project such as OCR and handling unexpected artifacts such as coughing during an ultrasound [4].
- ISO 14971 defines rules and a framework for risk management in medical devices, including SaMDs [5].
- PIPEDA and PHIPA are privacy regulations in Canada and Ontario for biometric data, where ECG signals combined with carotid ultrasound data could be considered as identifiable health records [6], [7].
- IEEE 754 is the standard on floating point arithmetic. By converting continuous health data to discrete floating-point numbers for numerical analysis, the standard is relevant for preventing floating point inaccuracies that may appear in the analysis [8].
- Open-Source Software (OSS) Licenses are relevant as common image processing, and numerical analysis packages have varying levels of permissiveness in their licenses. These must be analyzed as packages are chosen to ensure that the license requirements are suitable for the requirements of this project [9].

The consequences of non-compliance for SOR are that if the software was used in research, publishing it in a clinical journal would require full compliance. Similarly, commercial distribution is impossible without compliance. For the IEC and ISO standards, the risks are in any regulatory audits. Additionally, there is the risk of the software being created unable to handle medical edge cases and failure modes that the standards and frameworks would have ensured were accounted for.

Ethical Concerns

Relevant ethical concerns surround the identifiability of data and risks in using this software for medical diagnostics. For example, in using automated algorithms for many of the checks, the analysis moves from humans to machine. This has two effects: the human reviewer loses their skills in checking the data as they rely on automation more, and they trust its output more over

time and are less likely to catch incorrect measurements. These inaccuracies can distort studies that rely on this software and lead to incorrect outcomes from said studies.

Another concern is the ethics around using a limited dataset. Humans come from a diverse pool of ages, bodies, and inner vessels. As the only data that can be used is data taken from members of the team, its applicability to a diverse group that may be studied by this software in the future is minimal. Thus, edge detection algorithms and other diagnostic results may be ill-prepared for future study populations, leading to potentially incorrect or skewed data. This can lead to inequitable health diagnostics for various groups or perhaps misdiagnoses of underrepresented populations.

Influence on Design Concept Generation and Evaluation

Design concept generation is largely unchanged at the macro-level, because most of the identified risks and constraints involve lower-level decisions like which software packages should be used, or how to evaluate diagnostic data with minimal floating-point errors. Evaluation is the most impacted part of the design process, as many medical device standards for SaMDs are tailored around the software lifecycle and risk management. Thus, design evaluation will aim to follow the frameworks in place within these standards and set up the resulting software as an audit-ready application that adheres to all requirements within them. Concretely, this involves following design control procedures, formal benchmarking against the legacy software this project is being built to replace, unit testing in the code, and formal risk management that catalogues every software hazard and designs to reduce these risks.

References

[1] McMaster University, "Week 2 Design Studio Slides," IBEHS 1P10 Course Material, Avenue to Learn. [Online]. Available: https://avenue.cllmcmaster.ca/d2l/le/lessons/783612/topics/5744589 (accessed Oct. 23, 2026).

[2] KUBS Engineering Inc., "2026-10-05_Taha_Abbasi_Hashemi_Meeting.md," Capstone-Design-History-File Repository, GitHub, Oct. 5, 2026. [Online]. Available: https://github.com/KUBS-Engineering-Inc/Capstone-Design-History-File/blob/ec6fe89138df886e47b6d1d838a31df69706b87f/02_Meetings/Stakeholder_Meetings/2026-10-05_Taha_Abbasi_Hashemi_Meeting.md (accessed Oct. 23, 2026).

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[4] International Electrotechnical Commission, *Medical device software — Software life cycle processes*, IEC 62304:2006+AMD1:2015, 2015. [Online]. Available: https://www.iso.org/standard/64686.html (accessed Oct. 23, 2026).

[5] International Organization for Standardization, *Medical devices — Application of risk management to medical devices*, ISO 14971:2019, 2019. [Online]. Available: https://www.iso.org/standard/72704.html (accessed Oct. 23, 2026).

[6] Government of Canada, *Personal Information Protection and Electronic Documents Act*, S.C. 2000, c. 5, 2000. [Online]. Available: https://laws-lois.justice.gc.ca/eng/acts/P-8.6/ (accessed Oct. 23, 2026).

[7] Government of Ontario, *Personal Health Information Protection Act, 2004*, S.O. 2004, c. 3, Sched. A, 2004. [Online]. Available: https://www.ontario.ca/laws/statute/04p03 (accessed Oct. 23, 2026).

[8] IEEE, *IEEE Standard for Floating-Point Arithmetic*, IEEE Std 754-2019, 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8766229 (accessed Oct. 25, 2026).

[9] Open Source Initiative, "Licenses by Name," Open Source Initiative. [Online]. Available: https://opensource.org/licenses/ (accessed Oct. 23, 2026).

Appendix A

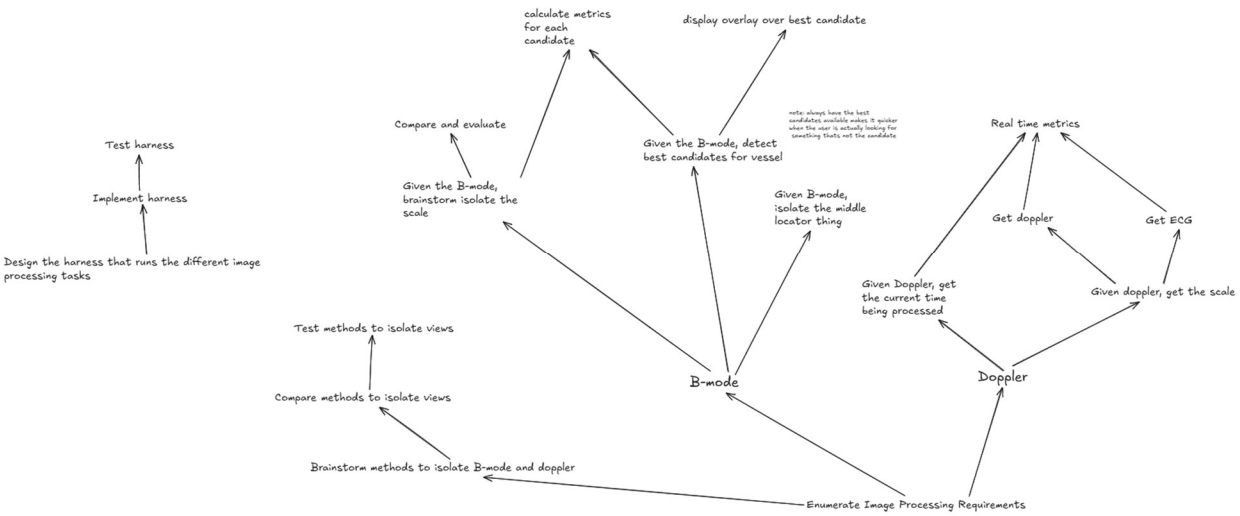


Figure A1. Breakdown of Tasks and Their Subcomponents for Research and Planning (Fall Term); Note that arrows indicate how a task is further broken down

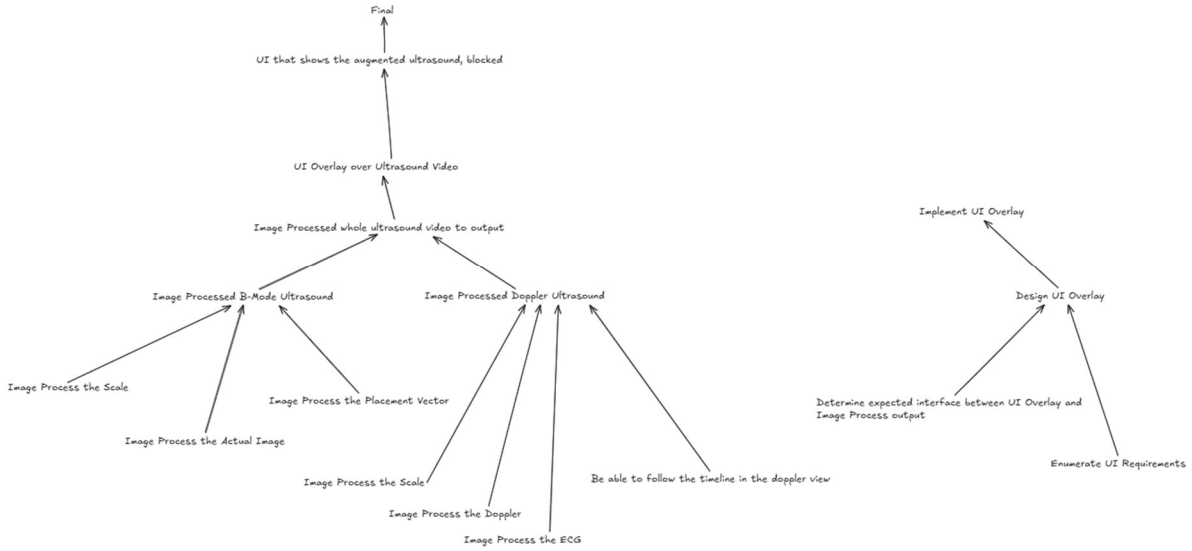


Figure A2. Breakdown of Tasks Implementation and Their Subcomponents (Winter Term); Note that arrows what tasks come together to create the final product (opposite way than Figure 1.)